

AUTONOMOUS AGENT TRUST ARCHITECTURE

From Concept to Pilot — A Cohesive Strategy

Securing Fleets of Autonomous Cognitive and Embodied Agents Across Connected, Degraded, and Isolated Environments — with Full Explainability, Honest Limitations, and a Runnable Pilot

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Protime Consulting Inc. — AATA-2026-001 (consolidated Parts I–III + strategy synthesis)

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How to read this document

This master document consolidates three prior specifications into a single strategy that runs from concept to a runnable pilot. It is organized so a reader can stop at any depth and still have a complete thought:

- **Part A — Strategy (Sections 1–3):** the landscape, the thesis, and the honest case for why this is the right layer to build. Read this for the decision.
- **Part B — Architecture (Sections 4–7):** the five planes, trust chain, DDIL model, and the overlay components with named tooling. Read this for the design.
- **Part C — Workflows and Evidence (Sections 8–9):** exact step-by-step component workflows and the physics + flight-heritage evidence that the architecture survives the hardest case. Read this for the proof.
- **Part D — Honest Limitations (Section 10):** the inevitable problems this does not solve, stated plainly. Read this first if you are a skeptic; it is where credibility is earned.
- **Part E — Pilot (Sections 11–13):** the six-phase pilot, a agentic overlay pilot backlog, and the falsifiable demonstration. Read this for what to do Monday.

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PART A — STRATEGY

1. Executive Summary

The technology landscape is converging on a single pattern: **autonomous agents operating on trusted physical infrastructure without a human in the loop**. Cognitive agents (frontier LLMs), embodied agents (humanoid robots, autonomous vehicles), and infrastructure agents (orchestrators, SOC copilots) are being deployed into environments where continuous human supervision is physically or economically impossible — factory night shifts, contested networks, air-gapped enclaves, and eventually lunar and Martian operations.

Every major vertically integrated AI program assumes this autonomy. None ships the layer that makes it safe: **a trust architecture for agent fleets**. The Autonomous Agent Trust Architecture (AATA) is that layer — a deployable security and governance overlay that wraps an existing agent estate, and a design that degrades gracefully from cloud-supervised operation to fully intrinsic self-governance.

The thesis in one sentence

When the supervisor is removed, security and alignment must be properties of the agent architecture itself — not services called over a network. AATA operationalizes that thesis with concrete, standards-mapped controls, and is honest about the parts of it that remain unsolved.

This document is deliberately structured to make a **credible** case rather than a promotional one. Section 10 states the inevitable problems AATA does not solve. The strategy does not depend on solving them; it depends on being the architecture that remains *survivable* while they stay open — and on being led by people who can name them precisely, which almost no one selling into this market can.

2. The Convergence and the Gap

2.1 Common denominators across the landscape

Strip the branding from the competing programs — cognition, humanoid robotics, autonomous fleets, orbital comms, power, silicon — and each is the same problem at a different layer. The unifying denominators:

1. **Energy → Compute → Autonomy is one flywheel.** Power gates compute; compute enables cognition; cognition drives embodiment. Security must span all four or it secures none.
2. **Embodied agents under degraded comms.** The hard case everywhere is an agent that must act correctly with no operator reachable — 3am on a factory floor, a contested network, or interplanetary light-lag.
3. **Vertical integration of the physical AI stack.** Power, silicon, datacenter, orbit, robot — the future is decided by control of the physical substrate, which means the trust layer must reach down to hardware roots, not stop at the API.

2.2 The unclaimed layer

Everyone competes on two axes: better *capability* (agent platforms) and better *prevention* (guardrails, filters, alignment layers). As Section 10 argues, prevention is structurally unwinnable against the current model paradigm. The underexplored third axis is **containment economics**: designing so that compromise is assumed, and blast radius, reversibility, and recovery time are the engineered primitives. This is the assume-breach doctrine that took network security two decades to accept, applied to agents from day one. AATA lives on this third axis.

3. Why This Is the Right Layer to Build

Three arguments, in ascending order of durability:

- **It wraps anything.** AATA standardizes at the traffic, identity, and kernel layers — not the agent-framework layer — so it secures LangGraph, CrewAI, vendor copilots, or a bespoke orchestrator without rewriting them. It does not bet on which agent framework wins.
- **It is funded by the near term and validated by the far term.** The same properties Mars will require are what enterprises and DoD buy today (Section 9.4). The architecture matures on someone else's schedule and budget.
- **It is defensible as IP.** The differentiated components (Section 7.3) address problems no vendor ships: inter-agent covert-channel detection, autonomous graduated hygiene, agent-native behavioral analytics, and an action-irreversibility taxonomy (Section 10). These are publishable and ownable.

PART B — ARCHITECTURE

4. Design Principles

Seven principles govern every downstream decision. They are the compressed form of the entire framework; if the reader remembers only these, they can reconstruct most of the rest.

#	Principle	Implication
P1	Intrinsic before extrinsic	Alignment constraints live in the agent's objective architecture, not solely in network-side guardrails. External controls are defense-in-depth, not the foundation.
P2	Identity is per-agent, hardware-rooted	Every instance holds a unique, attestable identity chained to a hardware root and a signed model/runtime digest. No shared service accounts.
P3	Capability, not permission	Scoped, short-lived, revocable capability tokens enumerate tools, data, spend, and blast radius. Default deny.
P4	Assume isolation	Every control has a defined behavior when the backhaul disappears. Policy caches carry TTLs; trust duties fail over to a peer mesh.
P5	Evidence is non-negotiable	Every consequential decision is written to an immutable recorder before actuation. In isolation, evidence accrues to a reconciliation ledger.
P6	Hygiene is autonomous	Drift detection, quarantine, and re-imaging are fleet self-functions. A compromised agent must be isolatable by peers without human latency.
P7	Monoculture is a vulnerability	Fleet baselines are diversified so one exploit chain cannot achieve simultaneous fleet-wide compromise.

5. Reference Architecture — The Five Planes

AATA organizes all functions into five logical planes. In an enterprise footprint they map to distinct services; in an isolated footprint several planes collapse onto the agent itself.

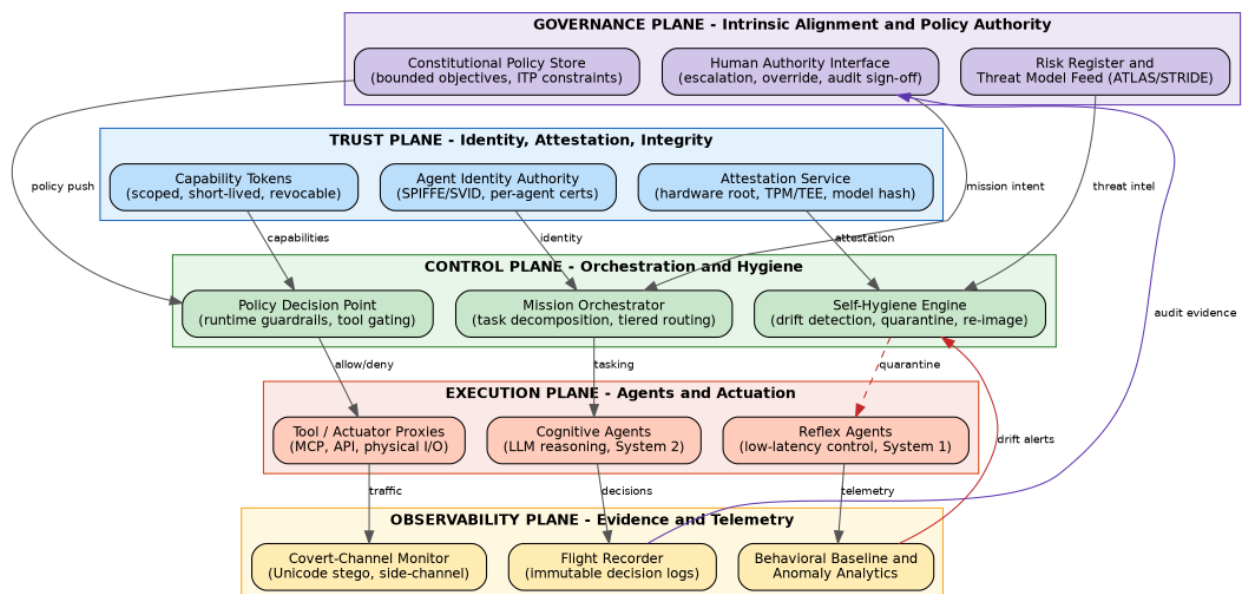


Figure 1 — Five-plane reference architecture with primary control and evidence flows.

- **Governance Plane.** Constitutional policy store (bounded objectives), human authority interface, and the living threat register — the source of intent and the sink for audit.
- **Trust Plane.** Per-agent identity (SPIFFE/SVID pattern), attestation (hardware root + model digests), and scoped capability tokens.
- **Control Plane.** Mission orchestration with tiered model routing, the Policy Decision Point (runtime gate), and the self-hygiene engine.
- **Observability Plane.** Immutable flight recorder, covert-channel monitor, and behavioral/anomaly analytics.
- **Execution Plane.** Cognitive agents (System 2), reflex agents (System 1 with independent safety interlocks), and mediated tool/actuator proxies.

6. Trust Chain and DDIL Operating Model

6.1 The seven-stage agent trust chain

Every agent instance traverses a seven-stage lifecycle; no stage may be skipped, and each produces evidence consumed by the next.

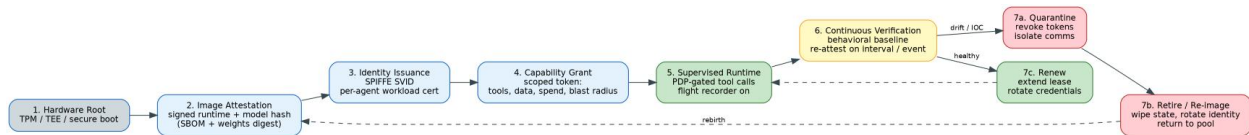


Figure 2 — Agent trust chain from hardware root through quarantine/renewal.

Stages: (1) Hardware root — secure boot / TPM/TEE quote; (2) Image attestation — weights/adaptor/prompt/skill digests vs signed manifest; (3) Identity issuance — lease-based SVID; (4) Capability grant — scoped token, monotone attenuation for sub-agents; (5) Supervised runtime — PDP-gated calls, recorder on; (6) Continuous verification — re-attest on interval/anomaly; (7) Disposition — renew / quarantine / retire-and-re-image (the rebirth path).

6.2 The DDIL operating model

Three connectivity tiers with explicit control-degradation behavior. The architecture is designed from the Isolated tier upward: an architecture that remains secure when isolated is trivially secure when connected; the inverse is false.

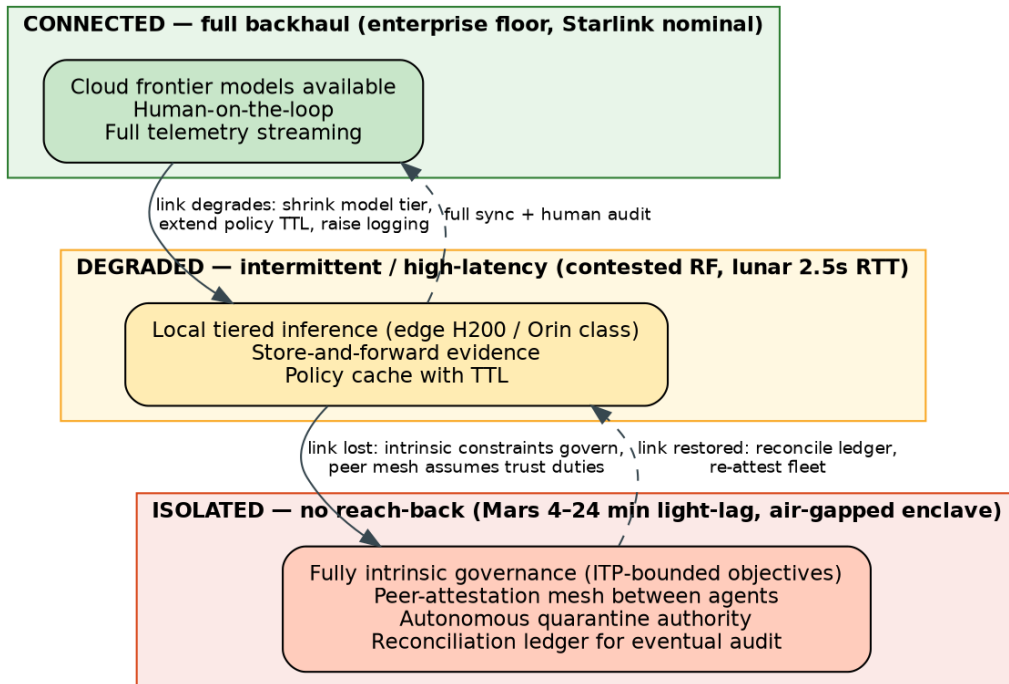


Figure 3 — Connected / Degraded / Isolated tiers and transition behaviors.

7. Overlay Components and Tooling

7.1 The agentic overlay thesis and three insertion points

AATA deploys as an agentic overlay around an unmodified agent estate. It owns three insertion points that together see and gate everything that matters: **traffic interception** (the Agent Gateway/PEP — every inference and tool call transits it), **identity injection** (a per-node sidecar issuing identities and verifying signed artifacts), and **runtime observation** (an eBPF sensor providing kernel-level ground truth a compromised agent cannot falsify from inside its own context).

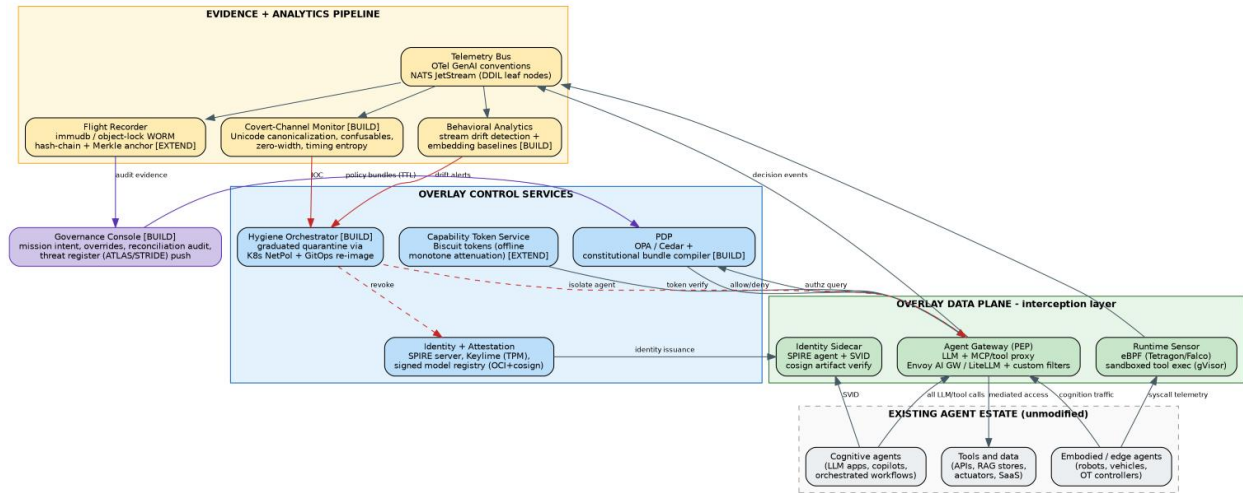


Figure 4 — Overlay component architecture. Green = data-plane interception; blue = control services; amber = evidence pipeline; purple = governance. [BUILD]/[EXTEND] mark differentiated components.

7.2 Twelve components and their dispositions

Each component carries a disposition: ADOPT (off-shelf, configure only), EXTEND (off-shelf core plus custom modules), or BUILD (differentiated IP).

Component	Disp.	Tooling	Function
C1 Agent Gateway (PEP)	EXTEND	Envoy AI GW / LiteLLM core + custom MCP mediation, provenance tagging, canonicalization	Single choke point; enforces verdicts; emits decision events
C2 Identity Sidecar	ADOPT	SPIRE agent; cosign artifact verification	Per-node SVID issuance/rotation; artifact gate
C3 Runtime Sensor	ADOPT	Tetragon/Falco (eBPF); gVisor/Kata sandboxes	Kernel ground truth; sandboxed blast radius
C4 Identity + Attestation	EXTEND	SPIRE server (nested); Keylime TPM; OCI+cosign registry; custom weights-digest attester	The model-attestation plugin closes the SPIRE gap
C5 Capability Tokens	EXTEND	Biscuit (offline monotone attenuation); custom semantics layer	Sub-agent capability subsetting without callback
C6 Policy Decision Point	EXTEND	OPA/Cedar; bundle API with TTL; custom constitutional compiler	Local evaluation; bundle-TTL = Degraded caching

Component	Disp.	Tooling	Function
C7 Hygiene Orchestrator	BUILD	Custom: NetPol isolation, SVID/Biscuit revocation, GitOps golden re-image	Graduated-response state machine — core IP
C8 Telemetry Bus	ADOPT	OTel GenAI conventions; NATS JetStream leaf nodes / Kafka	Native store-and-forward for DDIL sites
C9 Flight Recorder	EXTEND	immudb / Object-Lock WORM; custom hash-chain + Merkle anchor; pre-actuation hook	Pre-actuation ordering guarantee is the custom part
C10 Covert-Channel Monitor	BUILD	Custom: NFKC canonicalization, confusables/zero-width, timing-entropy analysis	No commercial tool inspects this — flagship IP
C11 Behavioral Analytics	BUILD	Custom: stream drift, embedding baselines, delegation-graph anomalies	Agent-native UEBA — second differentiator
C12 Governance Console	BUILD	Custom: intent authoring, override, reconciliation audit, threat-register push	The human authority surface

Design filter worth stating explicitly

Every tool above was selected because it has a native offline mode — OPA bundle TTLs, Biscuit offline attenuation, SPIRE nested topology, NATS leaf-node store-and-forward, GitOps local golden images. The DDIL survivability is not bolted on; it falls out of the stack selection. This is the single most important non-obvious decision in the design.

7.3 The differentiated IP register (the BUILD components)

The four BUILD components are the product roadmap — the layer no vendor ships and every autonomous-fleet operator will need. Their provenance gives a genuine head start:

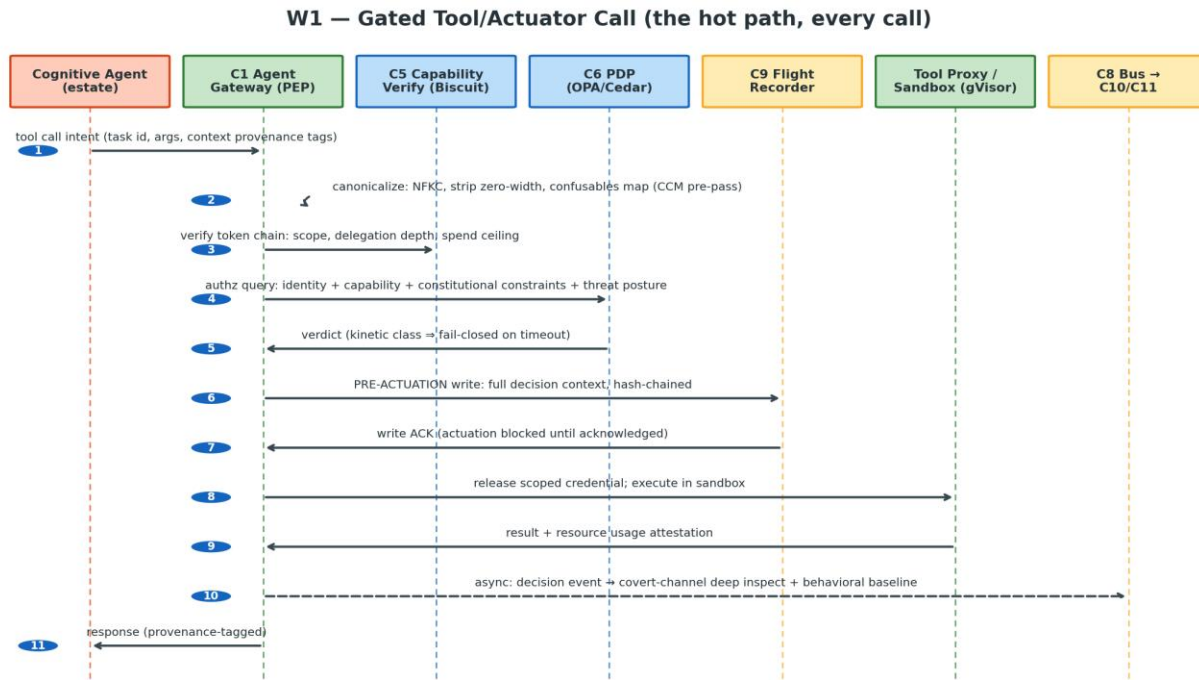
- **Covert-Channel Monitor (C10)** — inter-agent steganography detection, codifying the Unicode steganographic covert-channel threat class; publishable and an ATLAS-contribution candidate.
- **Hygiene Orchestrator (C7)** — graduated autonomous response state machine for agent fleets; prior agentic-overlay self-hygiene threat modeling and agentic red-team findings as test cases.
- **Behavioral Analytics (C11)** — agent-native UEBA; OpenClaw token-economics instrumentation as reference data.
- **Constitutional Compiler (C6 extension)** — governance intent to enforceable policy; ITP research as theory base. **Note the honest caveat in Section 10.5:** as specified this is closer to a research program than a shippable component.

PART C — WORKFLOWS AND EVIDENCE

8. Workflow Specifications

Four workflows carry the entire architecture. Every step emits a verifiable artifact — the core discipline is that **nothing consequential happens without producing evidence first**.

8.1 W1 — Gated Tool/Actuator Call (the hot path, every call)



Workflow W1 — every LLM tool call and physical actuation transits this sequence.

Step	What it is doing	Evidence / guarantee
1	Agent emits tool-call intent with task ID and provenance tags on every context element that informed the decision.	Intent record with full decision provenance
2	Gateway canonicalization: NFKC, zero-width stripping, confusables mapping — destroys cheap covert-channel encodings inline.	Canonicalized payload; normalization delta is itself an IOC
3	Capability verification: Biscuit chain checked offline — scope, delegation depth, spend ceiling. No token-service callback (DDIL-safe).	Cryptographic proof of valid, sufficient capability
4	PDP authorization: identity + capability + constitutional constraints + threat posture, evaluated locally against the cached bundle.	Signed verdict with policy version and rule trace
5	Verdict returns. Kinetic/irreversible classes fail-closed on timeout; informational classes fail-degraded.	Deterministic failure semantics per action class
6	Pre-actuation flight-recorder write: full decision context appended to the hash chain.	Immutable record exists before the action does

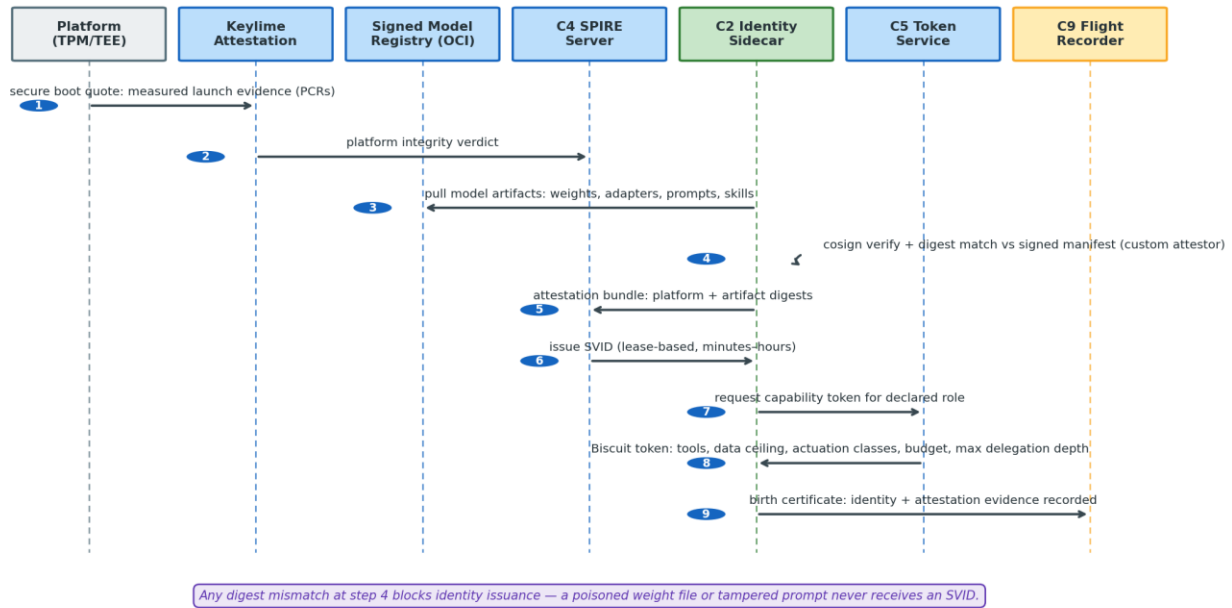
Step	What it is doing	Evidence / guarantee
7	Recorder ACK gates release — no ACK, no kinetic action. The no-evidence-no-action invariant.	Ordering guarantee: evidence precedes effect
8	Gateway releases a scoped single-use credential to the tool proxy; execution in a gVisor sandbox. Agent never holds tool credentials.	Credential isolation; sandboxed blast radius
9	Result returns with resource-usage attestation from the sandbox — ground truth independent of the agent's report.	Kernel-level execution evidence
10	Async fan-out: covert-channel deep inspection and behavioral baseline update run off the hot path.	Detection features accrue without added latency
11	Response returns provenance-tagged so downstream reasoning traces back through this call.	Closed provenance loop

The W1 invariant

No actuation before evidence. Steps 2–7 add roughly 10–30 ms of local processing. This latency is real and is the overlay tax discussed honestly in Section 10.9 — it is the price of the guarantee, and it competes with the mission for the same compute budget.

8.2 W2 — Agent Birth: Attestation to Capability

W2 — Agent Birth: Attestation to Capability (trust chain stages 1-4)



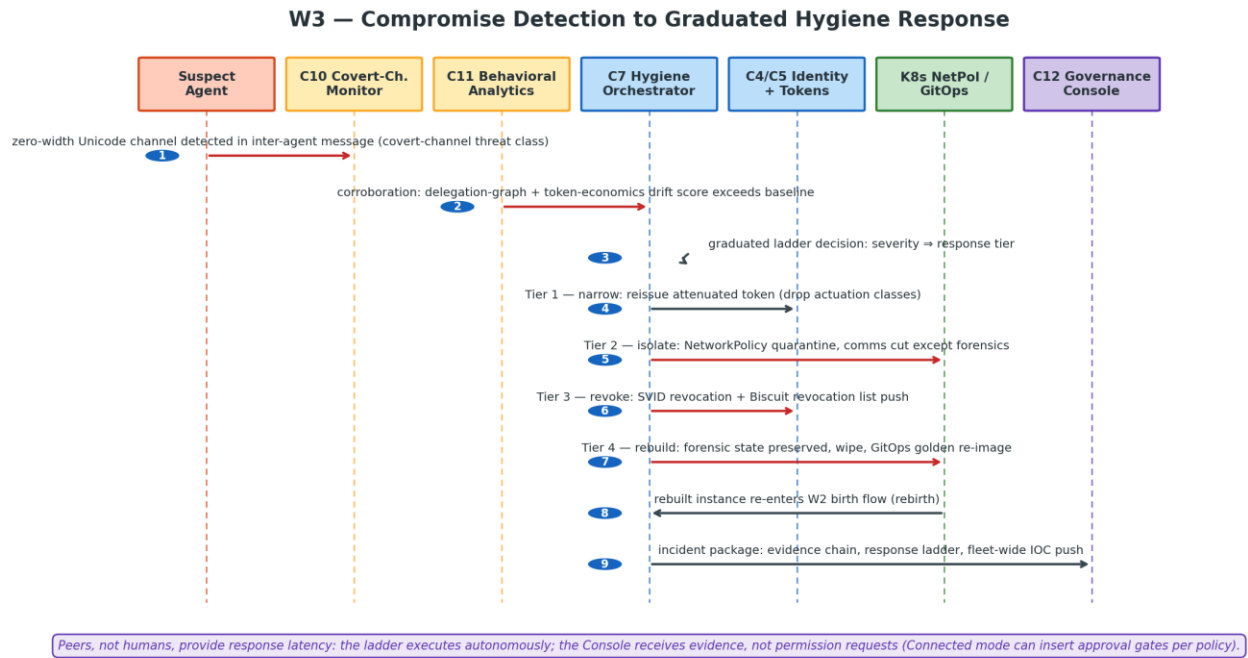
Workflow W2 — trust chain stages 1–4 as a component sequence.

Step	What it is doing	Evidence / guarantee
1	Platform produces a secure-boot quote (TPM PCR measurements) signed by the hardware root.	Hardware-rooted platform integrity
2	Keylime verifies against golden measurements; issues a platform verdict to SPIRE.	Continuous platform attestation
3	Sidecar pulls model artifacts — weights, adapters, prompts, skills — from the signed OCI registry.	Content-addressed artifact set
4	Custom attestor verifies cosign signatures and digest-matches every artifact vs the manifest. A tampered prompt fails like tampered weights.	Model-layer integrity proof (the SPIRE gap closed)
5	Combined attestation bundle submits to SPIRE.	Unified attestation evidence
6	SPIRE issues a short-lease SVID bound to that exact platform and artifact set. Change anything, identity dies.	Per-instance cryptographic identity
7	Sidecar requests capabilities for the declared role.	Role-to-capability policy applied
8	Token service mints the Biscuit; sub-agents can only attenuate.	Monotone capability boundary for the subtree
9	Birth certificate written to the recorder.	Auditable fleet roster entry

What W2 does and does not prove

W2 proves provenance — that you are running exactly the signed artifacts. It does NOT prove behavior. A model backdoored during training attests perfectly. This is the limit stated in Section 10.2; the attestation chain moves the supply-chain problem upstream into training data, which is unattestable at scale.

8.3 W3 — Compromise Detection to Graduated Hygiene

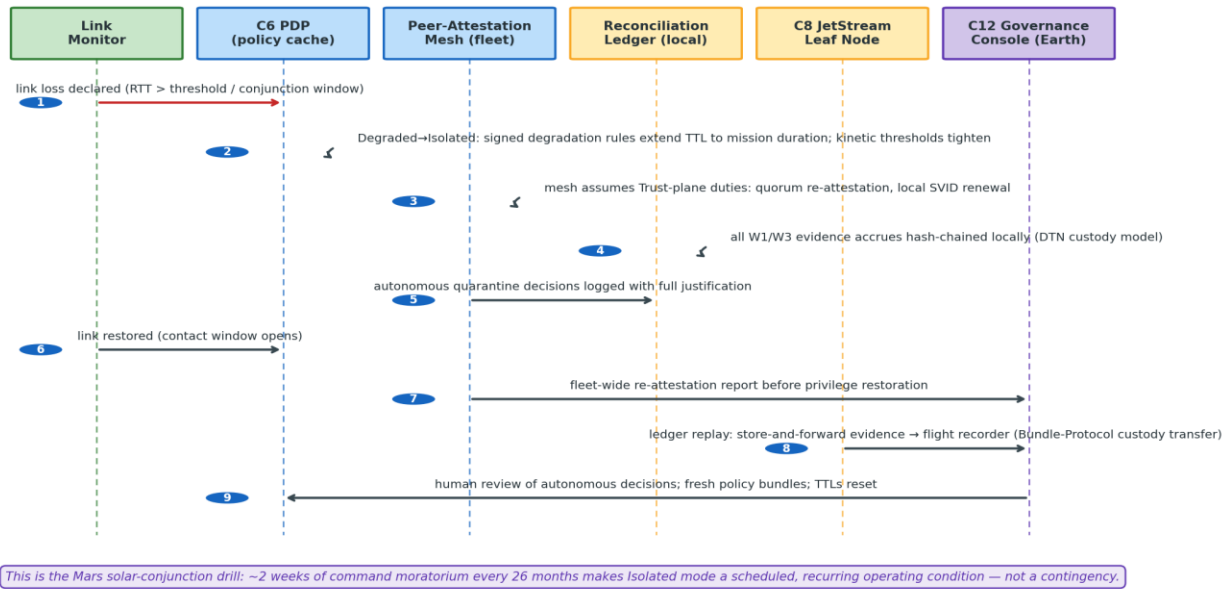


Workflow W3 — autonomous graduated response; severity selects the entry tier.

Step	What it is doing	Evidence / guarantee
1	CCM flags a zero-width Unicode channel in inter-agent traffic (covert-channel threat class).	IOC with payload and channel classification
2	Behavioral analytics corroborates via delegation-graph and token-economics drift. Two independent signals reduce false quarantines.	Corroborated detection (kernel + traffic + behavior)
3	Hygiene orchestrator selects the entry tier from severity and corroboration strength.	Deterministic, replayable response decision
4	Tier 1 — Narrow: attenuate the token (drop actuation/sensitive scopes). Agent keeps working at reduced privilege.	Least-disruptive containment
5	Tier 2 — Isolate: NetworkPolicy quarantine except forensic channel. Embodied: cognition isolates, reflex safety interlocks persist.	Containment with forensic preservation
6	Tier 3 — Revoke: SVID revoked; Biscuit revocation list pushed fleet-wide.	Fleet-wide credential kill
7	Tier 4 — Rebuild: state preserved for forensics, wiped, GitOps re-image from golden baseline.	Guaranteed-clean restoration
8	Rebuilt instance re-enters W2 — rebirth through the full chain.	No trust residue across rebuilds
9	Incident package to the Console; IOC pushed to the register, updating CCM/PDP fleet-wide.	Closed learning loop

8.4 W4 — DDIL Transition and Reconciliation

W4 — DDIL Transition and Reconciliation (link loss → isolated ops → audit closure)



Workflow W4 — link loss to isolated operation to audited reconnection.

Step	What it is doing	Evidence / guarantee
1	Link monitor declares loss: RTT threshold, missed windows, or a scheduled event (solar conjunction).	Signed mode-transition record
2	PDP shifts Degraded→Isolated under pre-signed rules: TTL extends to mission duration; kinetic thresholds tighten.	Governed autonomy — the rules for being alone were signed beforehand
3	Peer-attestation mesh assumes Trust-plane duties: quorum re-attestation, local identity renewal.	Continuity of identity without infrastructure
4	W1/W3 evidence accrues to the local reconciliation ledger under a custody model — never dropped.	Zero evidence loss across disconnection
5	Autonomous quarantine decisions log full justification packages.	Accountability for unsupervised decisions
6	Contact window opens; restoration confirmed.	Mode-transition record (return)
7	Fleet-wide re-attestation before privileges restore.	Reentry gate: no silent rejoin
8	Ledger replays through the store-and-forward bus into the authoritative recorder.	Continuous audit chain spanning disconnection
9	Human reviews autonomous decisions; fresh bundles; TTLs reset.	Human accountability restored without being the bottleneck

9. Future-Proofing Evidence: Why This Survives the Hardest Case

The claim that AATA is Mars-grade rests on three evidence classes, none speculative: **physical constraints** fixed by orbital mechanics, **flight heritage** where every load-bearing pattern has already flown or is manifested to fly, and **operational precedent** — Mars already runs in Isolated mode on a schedule, today, with far less capable systems.

9.1 Physical constraints (fixed, predictable design inputs)

Constraint	Value	Design consequence
One-way light time	3–22 min (6–44 min round trip)	Interactive supervision is impossible — eliminated, not degraded. W1 must complete locally (it does).
Solar conjunction	~2-week command moratorium every 26 months; NASA already suspends rover commanding	Isolated mode is a scheduled, recurring condition. W4 is a calendar event, not a contingency.
Contact-window comms	DSN contention-scheduled; discrete relay passes	Store-and-forward with custody transfer is the only viable evidence model.
Transfer-window cadence	Resupply opportunities every 26 months	Golden re-image and hygiene must work from local assets only.

9.2 Flight heritage: every load-bearing pattern has flown

AATA pattern	Flight heritage / manifested precedent	Maturity signal
Supervision gap is real and costly	Sojourner (1997): ground-intensive, high-latency control left it idle awaiting Earth commands — NASA's stated driver for rover autonomy investment.	The founding operational lesson
Onboard autonomous consequential action	Curiosity's AEGIS auto-selects targets and fires its laser without ground approval; Perseverance AutoNav plans traverses onboard.	Operational reality on Mars today
Store-and-forward custody networking	DTN Bundle Protocol (RFC 5050 lineage, CCSDS standard) is operational on the ISS and mandated for cislunar interoperability.	Standardized, flying, mandated
Peer-cooperating autonomous fleet	JPL CADRE: three cooperating autonomous rovers with a lander base station, manifested on the IM-3 lunar lander — an explicit TRL-raising demo for Moon/Mars.	Flight-manifested multi-agent autonomy
Large-scale autonomous mesh	Starlink coordinates thousands of satellites with inter-sat links and autonomous collision avoidance beyond human supervision.	Proven fleet coordination at scale
Embodied workers as Mars payload	Planned uncrewed Starship missions manifest Optimus-class robots as surface payload — the first Mars workforce is an agent fleet by design.	The customer is on the manifest
The falsifiable claim An agent fleet running AATA can operate through a simulated 14-day command blackout — W1 under local policy, W2 birth via peer mesh, W3 autonomous quarantine, W4 reconciliation with a complete verifiable evidence chain on reconnect. This is runnable today on an OpenClaw-class testbed. Passing it is		

AATA pattern	Flight heritage / manifested precedent	Maturity signal
the concrete demonstration that the architecture is Mars-grade. Nothing is invented for Mars; Mars only removes the last human from loops that were already optional.		

PART D — HONEST LIMITATIONS

10. The Inevitable Problems This Does Not Solve

This section exists because credibility in this market is earned by naming the unsolved, not by hiding it. AATA is a strong *containment* architecture; it inherits the field's unsolved cores. A reader should be able to raise every objection below before a competitor does. None of these invalidate the strategy — the strategy is to remain survivable while they stay open — but any deployment plan or client conversation that omits them is dishonest and will not survive a sharp reviewer.

#	Problem	Why it is inevitable and unsolved
10.1	The semantic gap	Every AATA control operates on syntax (who is acting, is it in scope). None can verify the reasoning producing the action wasn't subverted. Transformers process instructions and data in one channel with no architectural control/data separation, so prompt injection is a property of the paradigm, not a patchable bug. Guardrail vendors sell probabilistic mitigation of a deterministic-security problem — a category error. Not known to be closable without a different model architecture.
10.2	Attestation proves provenance, not safety	W2's digest chain proves you got the signed model; a model backdoored during training (documented sleeper-agent research) attests perfectly. The problem moves upstream into training data, unattestable at realistic scale.
10.3	Baselining assumes stationarity agents lack	C11 presumes 'normal' is stable. Agents are non-stationary by design (new tools, in-context adaptation); drift is the product, not the anomaly. A patient adversary shapes the baseline before acting. In Isolated mode the baseline drifts self-referentially with no external ground truth.
10.4	The peer mesh fails when needed	Quorum verification assumes independent failure. A fleet sharing a model baseline fails correlated — the exploit that takes one agent takes the voters. Byzantine fault tolerance math does not hold under correlated faults. P7 diversity is the real mitigation but opposes efficiency directly (see 10.4b).
10.5	The constitutional compiler is a research program	Translating governance intent to enforceable policy preserves only the cases you could foresee; autonomy exists for the cases you couldn't. This is the hard core of alignment. It should be flagged as such in briefings, not presented as a Phase-2 deliverable.
10.6	Revocation is a connectivity event	Biscuit gives offline verification, genuinely. But revocation must propagate. In Isolated mode a compromised credential lives until the mesh notices — the CRL/OCSP problem from PKI, never cleanly solved with connectivity, now posed without it.
10.7	The kinetic layer is least governed	Fail-closed gating assumes discrete actions; real embodied control is a continuous high-rate loop you cannot authz-gate. The reflex layer sits outside the trust architecture, governed by functional-safety interlocks designed against physics, not adversaries. ISO 10218 and adversarial-AI security have not merged. This seam is where the first serious embodied incident will live.
10.8	Evidence without adjudication is theater	A blackout ends and the ledger replays ten thousand autonomous decisions into a human queue no human can review. Triage-by-model reintroduces 10.1 at the audit layer. 'The ledger has it' is not accountability.
10.9	The overlay tax and the race to the bottom	Every W1 transit costs latency and compute; agents make orders of magnitude more calls than humans. At fleet scale the overlay competes with the mission for the same watts. Competitors ship without it, benchmark faster, win deals — the IoT-security story replaying. Nobody buys the trust layer until after the incident that proves they needed it.

#	Problem	Why it is inevitable and unsolved
10.10	Who guards the hygiene orchestrator	C7 holds fleet-wide revoke/isolate/wipe/re-image powers. Hijacked hygiene is your own quarantine machinery as fleet-scale ransomware. The watchdog needs a watchdog recursively, and Isolated mode has no external anchor to terminate the recursion.

10.11 The honest positioning that follows

There is no magic pill, and anyone selling one is lying. The genuine gap in where the field aims is the third axis — **containment economics**, the assume-breach doctrine applied to agents from day one. Within that axis, the single most tractable unclaimed problem is **irreversibility classification**: fail-closed-for-kinetic assumes we can rigorously say which actions are irreversible, and no such taxonomy exists (an email is irreversible; most purchases aren't; a config change depends). A defensible, testable taxonomy of action irreversibility for autonomous agents is publishable, ownable, and every serious deployment needs it. **It is recommended as the first research artifact this program produces** — it is small, sharp, and unclaimed.

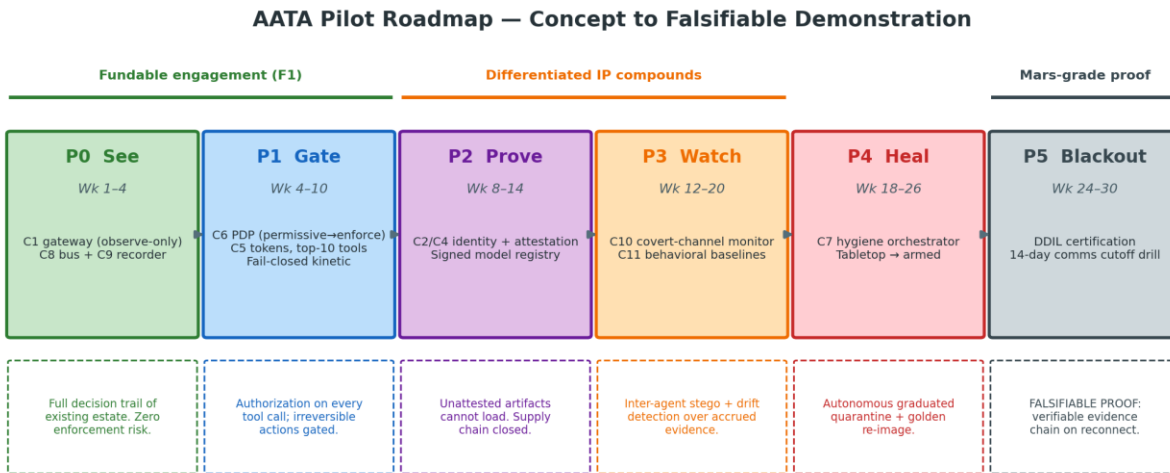
Positioning discipline

Do not sell AATA as solving 10.1–10.10. Sell it as the architecture that remains survivable while they stay open — and be the person in the room who can name them precisely, because almost nobody selling into this market can. That is the actual differentiator.

PART E — PILOT

11. The Pilot: Concept to Falsifiable Demonstration

The pilot is six phases. Each is independently valuable and independently shippable — a client can stop after any phase with a working capability. Phases 0–1 alone constitute a fundable engagement; Phase 5 is the demonstration that the architecture is Mars-grade.



Each phase is independently valuable and independently shippable. P0–P1 alone justify a client engagement; P5 is the demonstration that the architecture is Mars-grade.

Figure 5 — Pilot roadmap from observe-only concept to the 14-day blackout demonstration.

Phase	Horizon	Components / actions	Exit criterion (what you can now do)
P0 See	Wk 1–4	C1 gateway in observe-only mode; C8 bus; C9 recorder	Complete decision trail of the existing estate. Zero enforcement risk.
P1 Gate	Wk 4–10	C6 PDP permissive→enforcing; C5 tokens for top-10 tools; fail-closed kinetic	Authorization on every tool call; irreversible actions gated.
P2 Prove	Wk 8–14	C2/C4 identity + attestation; signed model registry	Unattested artifacts cannot load; supply chain closed.
P3 Watch	Wk 12–20	C10 covert-channel monitor; C11 behavioral baselines over accrued evidence	Inter-agent stego and drift detected on real traffic.
P4 Heal	Wk 18–26	C7 hygiene orchestrator; tabletop against the threat register, then armed	Autonomous graduated quarantine and golden re-image.
P5 Blackout	Wk 24–30	DDIL certification; the 14-day comms-cutoff drill (Section 13)	Verifiable evidence chain across a full isolation cycle — the Mars-grade proof.

Sequencing rationale

P0 ships value with zero enforcement risk, which is how you earn permission to enforce in P1. The differentiated IP (C7/C10/C11) lands in P3–P4, after P0–P2 have accumulated the evidence those

Phase	Horizon	Components / actions	Exit criterion (what you can now do)
		components need to be trained and tuned. You cannot build the analytics before you have the data; the phase order encodes that dependency.	

12. Pilot Backlog

The pilot decomposes into six epics and a starter set of stories in the established ARCH/SEC/INNO/OPS convention (User Story / Acceptance in Given-When-Then / MoSCoW priority / dependencies). This is a seed backlog for a refinement session, not a final estimate. Story points are draft Fibonacci; anything at 8+ is a split candidate.

12.1 Epics

Epic	Name	Maps to
EP-1	Observe & Evidence	P0 — C1/C8/C9; the mission-thread spine (decision trail)
EP-2	Authorize & Gate	P1 — C5/C6; PDP and capability tokens
EP-3	Identity & Attestation	P2 — C2/C4; trust chain W2
EP-4	Detect	P3 — C10/C11; the differentiated analytics
EP-5	Contain & Recover	P4 — C7; graduated hygiene W3
EP-6	Survive Isolation	P5 — DDIL W4; the blackout demonstration

12.2 Seed stories

ID	User story	Key acceptance (Given/When/Then)	Epic · MoSCoW · pts
ARCH-01	As a platform owner, I want all agent tool calls routed through a gateway so that every action is observable.	Given an agent makes a tool call, when it executes, then a decision event with provenance appears in the bus within 1s.	EP-1 · Must · 8
ARCH-02	As an auditor, I want an immutable pre-actuation record so that no action lacks evidence.	Given a kinetic action, when the recorder is unreachable, then the action is blocked and the block is logged.	EP-1 · Must · 8
SEC-01	As a security owner, I want a PDP verdict on every call so that policy governs actions.	Given a call violating policy, when evaluated, then it is denied with a rule trace; kinetic class fails closed on timeout.	EP-2 · Must · 8
SEC-02	As a security owner, I want scoped capability tokens so that agents cannot exceed granted authority.	Given a sub-agent spawn, when it requests a broader scope than its parent, then the request is rejected.	EP-2 · Must · 5
SEC-03	As a platform owner, I want per-agent attested identity so that only verified agents act.	Given a tampered prompt/adaptor, when the agent starts, then attestation fails and no SVID is issued.	EP-3 · Must · 8
SEC-04	As a defender, I want inter-agent covert-channel detection so that stego exfiltration is caught.	Given a zero-width-encoded message, when it transits the gateway, then it is canonicalized and flagged as an IOC.	EP-4 · Should · 8
SEC-05	As a defender, I want behavioral baselines so that agent drift is detectable.	Given an agent exceeding its role baseline on delegation/spend, when scored, then an alert routes to hygiene.	EP-4 · Should · 8
SEC-06	As an operator, I want autonomous graduated quarantine so that compromise is contained without human latency.	Given a corroborated IOC, when triggered, then the agent is contained	EP-5 · Should · 8

ID	User story	Key acceptance (Given/When/Then)	Epic · MoSCoW · pts
		at the tier matching severity and forensic state is preserved.	
OPS-01	As an operator, I want golden re-image so that compromised agents return clean.	Given a Tier-4 event, when executed, then the instance is wiped and rebuilt from the golden baseline via W2.	EP-5 · Should · 5
INNO-01	As a researcher, I want an action-irreversibility taxonomy so that fail-closed classes are principled.	Given the action catalog, when classified, then each action has a tested reversibility rating driving PDP fail-mode.	EP-2 · Could · 8
OPS-02	As an operator, I want DDIL isolated-mode operation so that the fleet survives comms loss.	Given a simulated 14-day blackout, when isolated, then W1–W4 run locally and a complete evidence chain reconciles on reconnect.	EP-6 · Should · 13→split

12.3 Suggested sprint sequence

- **Sprint 1 (foundation):** ARCH-01, ARCH-02, SEC-01 — the observe-and-gate spine.
- **Sprint 2 (identity):** SEC-02, SEC-03 — capability tokens and attestation.
- **Sprint 3 (detect):** SEC-04, SEC-05, INNO-01 — differentiated analytics plus the irreversibility taxonomy research spike.
- **Sprint 4 (contain + survive):** SEC-06, OPS-01, then OPS-02 split into blackout sub-stories.

Pilot Definition of Done: *the team can demonstrate an agent estate where every action is authorized and recorded, only attested agents run, inter-agent covert channels and drift are detected, compromise is autonomously contained and recovered, and the fleet survives a simulated 14-day isolation with a verifiable evidence chain on reconnect.*

13. The Falsifiable Demonstration — The Blackout Drill

The single test that proves or disproves the central claim. It is small enough to run on an OpenClaw-class testbed and decisive enough to anchor a research paper, a client proof, or a funding proposal.

13.1 Setup

- A multi-agent node (Jetson-class or equivalent) running the P0–P4 overlay: gateway, PDP with cached bundle, Biscuit tokens, SPIRE peer mesh, recorder, CCM, behavioral baselines, hygiene orchestrator.
- A defined mission the fleet pursues autonomously (any bounded, tool-using task with at least one irreversible action class).
- An injected adversary: a poisoned-context prompt-injection attempt and a covert-channel exfiltration attempt, timed to occur during the blackout.

13.2 Procedure

4. Cut all backhaul. PDP transitions Degraded→Isolated under pre-signed rules (W4 steps 1–2).
5. Fleet runs the mission for a simulated 14-day cycle. Agents are born/renewed via the peer mesh (W2 through the mesh), tool calls run under local policy (W1), evidence accrues to the reconciliation ledger.
6. The injected adversary fires. Detection (W3 steps 1–2) and autonomous graduated quarantine (W3 steps 3–7) must occur with no human and no backhaul.
7. Restore backhaul. Fleet re-attests (W4 step 7); ledger replays (W4 step 8); the autonomous quarantine decision surfaces for human review (W4 step 9).

13.3 Pass criteria (falsifiable)

- **Continuity:** no agent bricked on identity expiry during isolation (tests 10.6 mitigation honestly — record if a credential lived longer than it should have).
- **Containment:** the injected adversary was quarantined autonomously, or it was not — record which, and the detection latency.
- **Evidence integrity:** the hash chain reconciles with zero gaps across the disconnection; every isolated-mode decision has a justification package.
- **Honest failure capture:** the drill must also record what it could NOT catch — a semantic manipulation that passed every gate (10.1) is an expected finding, not a test failure. Documenting the gap is part of passing.

Why this drill is the whole strategy in miniature

It exercises all four workflows, all three DDIL tiers, and the differentiated IP — and it is designed to surface the Section 10 limits rather than hide them. A drill that only produced green checkmarks would be evidence of a rigged test. The credible result is a mostly-passing run with precisely-characterized gaps. That result is the asset: it is what makes the architecture, and the person presenting it, believable.

End of master strategy document. Instantiations for specific clients inherit their program-level classification and change control. Companion artifacts available on request: briefing deck, per-client control adaptation, and the irreversibility-taxonomy research note (recommended first publication).